

Pearson Edexcel

**Level 3 GCE — Predicted Paper — Hard
Economics A (9EC0/03)**

Paper 3: Microeconomics and Macroeconomics

Synoptic paper. Two case-study sections combining micro and macro.

MARK SCHEME

Total Marks: 100 • Time: 2 hours

(For revision and practice use only.)

General marking guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme — not according to their perception of where the grade boundaries may lie.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed-out work should be marked UNLESS the candidate has replaced it with an alternative response.

Hard-tier expectations

This paper targets high-performing A Level candidates. Level 5 responses should engage naturally with higher-order concepts (for example: hysteresis, Ricardian equivalence, expectations-augmented Phillips curve, crowding out, debt dynamics (g vs r), quantitative tightening, capital deepening). Simply defining a term will not secure top-band marks — the concept must be integrated into the chain of reasoning or evaluation.

Assessment Objectives — weighting in Paper 3

- AO1: Demonstrate knowledge of terms/concepts and theories.
- AO2: Apply knowledge to problems and issues in familiar and unfamiliar contexts.
- AO3: Analyse issues within economics, showing an understanding of their impact on economic agents.
- AO4: Evaluate economic arguments and use qualitative and quantitative evidence to support informed judgements relating to economic issues.

Levels-based marking for 25-mark questions

Indicative content is provided as a guide and is not exhaustive. Examiners should use the levels descriptors below to award an overall level, then use a best-fit approach to award a mark within the level.

Level	Marks	Descriptor
1	1–5	Isolated knowledge, largely descriptive. Limited use of the extract/figure. Very limited or no analysis; no evaluation.
2	6–10	Sound knowledge of at least two relevant concepts. Some relevant application. Analysis is developed but may be one-sided or superficial for a Hard-tier question. Evaluation is attempted but undeveloped.
3	11–15	Good, accurate knowledge across several concepts. Application is consistent and uses data/extract. Analysis shows logical chains and engages with at least one higher-order concept (e.g. hysteresis, Ricardian equivalence, expectations). Evaluation is attempted.
4	16–20	Strong, accurate knowledge integrating micro and macro. Application is precise and data-driven. Analysis is well-developed, two-sided, and engages critically with higher-order theory. Evaluation is explicit with supported judgements.

Level	Marks	Descriptor
5	21–25	Excellent, fully synoptic knowledge. Application is sophisticated and embedded. Analysis is rigorous, chained and balanced, integrating higher-order concepts naturally. Evaluation is prioritised, contextual and leads to a reasoned, supported judgement that recognises trade-offs and conditioning assumptions.

Question 1(a) — Calculation and comment (5 marks)

Simple average annual UK productivity growth, 2015–2025, with brief comment.

Marks breakdown

AO1	AO2	AO4
1	2	2

Expected working and answer

- Sum of annual values: $1.0 + 0.8 + 0.9 + 0.5 + 0.3 + (-1.2) + 1.5 + 0.1 + (-0.3) + 0.2 + 0.4 = 4.2$.
- Divide by 11 years: $4.2 / 11 \approx 0.38\%$ per year. (Accept 0.38% or 0.4%.)
- Comment: confirms extract description of productivity growth “barely 0.4% per year” — markedly below the ~2.2% pre-2008 trend.

Marking notes

- 1 mark for correct identification of values; 2 marks for correct method (sum/n); 2 marks for correct answer with comment.
- OFR applies to a minor arithmetic slip if method is sound.
- No comment caps at 4/5.

Question 1(b) — Discuss (12 marks)

Macroeconomic effects of sustained weak productivity growth on real wages and long-run inflation, with AD/AS diagram.

Marks breakdown

AO1 (K)	AO2 (App)	AO3 (An)	AO4 (Ev)
3	3	3	3

Indicative content

- Knowledge: productivity (output per hour) as a driver of LRAS; real wages $\approx$ productivity in the long run; unit labour costs = wage / productivity.
- Application: extract data — real wages barely higher than 2008; double-digit inflation in 2022 despite weak growth; weak capital deepening.
- Diagram: LRAS stagnates (shifts less than AD), widening inflationary gap at each level of output; show SRAS and AD intersecting LRAS.
- Analysis (real wages): slow productivity growth $\rightarrow$ flat real wages $\rightarrow$ living standards stagnation $\rightarrow$ political pressure for nominal wage increases not justified by productivity.
- Analysis (inflation): rising unit labour costs $\rightarrow$ cost-push pressure; weak LRAS means demand stimulus feeds into prices rather than output; persistent output gap is smaller $\rightarrow$ higher NAIRU.

Evaluation points expected

- Depends on source of low productivity — cyclical (post-pandemic scarring) vs structural (skills, capital, management).
- Expectations — wage-price spirals more likely when inflation expectations de-anchor.
- Short-run vs long-run — in the short run, real wages can diverge from productivity.
- Compositional effects — sectoral shifts (services expanding) may bias measured productivity.
- Possible mitigants — migration, investment, competition policy — can re-raise productivity.

Marking notes

- Correct AD/AS diagram is essential; cap at 9/12 without a properly labelled diagram.
- Reward candidates linking productivity explicitly to unit labour costs and to a shift (or failure to shift) of LRAS.

Question 1(c) — Examine (8 marks)

Advantages of public investment in infrastructure and R&D over cuts in corporation tax as a productivity-boosting tool.

Marks breakdown

AO1 (K)	AO2 (App)	AO3 (An)
2	2	4

Indicative content

- Directly addresses market failures: underprovision of R&D and public goods (infrastructure) due to positive externalities.
- Raises the marginal product of private capital (public–private capital complementarity), crowding in private investment.
- Corporation tax cuts are poorly targeted — benefit unproductive incumbents as well as productive scale-ups; deadweight loss if firms would have invested anyway.
- Corporate tax changes are subject to profit-shifting and international competition; investment incentives are more geographically sticky.
- R&D tax credits (referenced in extract) are a hybrid — more targeted than headline rate cuts.
- Fiscal multiplier of public investment typically > fiscal multiplier of corporation tax cuts in credible empirical studies.

Application expected

- Extract references: “low public investment in infrastructure and R&D”, “full expensing of capital”, “R&D tax credits”.
- Figure reference: sub-1% productivity growth despite a slightly rising investment ratio — suggests composition, not just level, matters.

Marking notes

- “Examine” — no explicit requirement for a balanced judgement, but one-sided answers cap at upper Level 2.
- Top-band candidates should recognise that the appropriate choice depends on whether the binding constraint is supply of capital or its productivity.

Question 1(d) — Evaluate (25 marks)

Micro and macro factors for a UK government policy package to reverse productivity stagnation by 2030.

Marks breakdown

AO1 (K)	AO2 (App)	AO3 (An)	AO4 (Ev)
4	6	7	8

Indicative content — microeconomic factors

- Market failures: R&D externalities, network effects in infrastructure, skills under-investment, information asymmetries in SME finance.
- The “long tail” of low-productivity firms — competition policy, insolvency reform, management training (e.g. Made Smarter).
- Labour-market frictions: skills mismatches, apprenticeships, post-16 education, lifelong learning.
- Capital markets: scale-up finance gap, patient long-term capital, regulation of pension funds and insurers.
- Planning regime: commercial property, lab space, data-centre capacity — supply-side constraints on productive investment.
- Regulation: quality vs cost trade-offs, cumulative regulatory burden on SMEs.

Indicative content — macroeconomic factors

- Aggregate demand stability: erratic demand damages capital spending (uncertainty effects).
- Real interest rates and the cost of capital — interaction with monetary policy and gilt yields (link to Section B).
- Fiscal space: is public investment constrained by debt rules? Link to debt sustainability ($g > r$).
- Trade and openness: Brexit friction, trade agreements, FDI inflows — affect competitive pressures and technology diffusion.
- Exchange rate stability for investment planning.
- Macroeconomic credibility — policy consistency essential for long-lived investments.

Evaluation points expected

- Time horizon — productivity policies take 5–10+ years; political cycle creates commitment problems.
- Prioritisation — binding constraint may be skills in one sector, capital in another.
- Opportunity cost — public investment crowds out other spending unless growth-financed.
- Measurement issues — productivity in services (especially public services) is hard to measure.
- Ricardian equivalence considerations if funded by borrowing.
- Distributional consequences — regional (London/SE vs other regions) and across skills groups.
- International context — UK cannot insulate from global productivity trends.
- Best packages combine supply-side micro reforms with macro stability and patient funding.

Levels apply (see front of mark scheme)

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Question 1(e) — Evaluate (25 marks)

Micro and macro effects of a sustained rise in UK business investment to 20% of GDP by 2030.

Marks breakdown

AO1 (K)	AO2 (App)	AO3 (An)	AO4 (Ev)
4	6	7	8

Indicative content — microeconomic effects

- Capital deepening — higher capital per worker raises marginal product of labour and hence real wages.
- Firm-level productivity gains through embodied technical progress in new equipment.
- Reallocation effects — capital flows from low to high productivity firms, narrowing the “long tail”.
- Supply-chain effects — agglomeration, knowledge spillovers among investing firms.
- Labour market: demand for skilled workers rises, widening skill premium unless matched by training.
- Competition effects — new entrants and technology diffusion raise consumer welfare via lower prices and higher quality.

Indicative content — macroeconomic effects

- AD effects: higher investment → AD shifts right in the short run (multiplier effects on employment and output).
- LRAS effects: sustained investment raises productive capacity → LRAS shifts right, permanently raising potential output.
- Inflation: higher LRAS relieves capacity constraints, potentially lowering long-run equilibrium inflation for a given AD path.
- Real wages rise in line with productivity — reverses post-2008 stagnation.
- Balance of payments: initial deterioration (import content of capital goods) then potential improvement through export competitiveness.
- Fiscal position: higher GDP → higher tax revenues; debt-to-GDP ratio falls via denominator; effective real rate falls relative to growth (improves g vs r).
- Monetary policy: higher neutral real rate may allow Bank Rate to normalise without recession.

Diagram suggestions

- AD/AS with both short-run (AD right) and long-run (LRAS right) shifts.
- Production possibility frontier — outward shift.

Evaluation points expected

- Feasibility — moving from ~17% to 20% of GDP represents a structural break; is this politically and institutionally plausible?
- Composition matters as much as level — investment in intangibles, R&D and skills has higher productivity impact than plant and buildings.
- Time lags — most effects realised beyond 2030; J-curve in productivity as firms adapt.
- Crowding out — if funded by public borrowing; partial crowding-in if it raises the neutral rate with credible fiscal policy.
- Hysteresis in reverse — sustained investment could reverse labour force scarring.
- Exchange rate feedback — higher growth may appreciate sterling, hurting exports.
- Sensitivity to external conditions — global recession could frustrate the build-out.

- Likely significant net positive, but realistic evaluation should assess probability of achieving the target.

Levels apply (see front of mark scheme)

Question 2(a) — Explain (5 marks)

Relationship between UK public sector net debt and 10-year gilt yield, 2015–2025.

Marks breakdown

AO1 (K)	AO2 (App)
2	3

Indicative content

- 2015–2019: debt rose slowly (80.5% → 83.5%) while yields fell from 1.8% to 0.8% — inverse relationship, driven by QE and low global real rates.
- 2020–2021: debt jumped to ~95% on pandemic borrowing; yields remained ultra-low (0.2–0.8%) due to massive BoE QE.
- 2022–2025: debt plateaus near 98–99% but yields rise sharply to 3.3–4.3% — positive relationship as market prices in higher risk and the end of QE.
- Interpretation: the relationship is not mechanical. In 2015–2021 yields tracked global real rates and QE; from 2022, structural shifts (QT, elevated inflation risk, mini-budget) caused the positive linkage to reassert.

Marking notes

- 1–2 marks: simple description of movements.
- 3–4 marks: identifies the regime shift around 2022 with data reference.
- 5 marks: links both periods to underlying drivers (QE, QT, global rates, risk premium).

Question 2(b) — Examine (8 marks)

Effects on UK government finances of the rise in the 10-year gilt yield from 0.8% to 4.3%, with bond price / yield diagram.

Marks breakdown

AO1 (K)	AO2 (App)	AO3 (An)
2	2	4

Indicative content

- Higher yields raise the cost of new borrowing and of refinancing maturing debt.
- With debt near 99% of GDP, a ~3.5 percentage point rise in yields produces a substantial rise in debt interest on rolled-over and new issuance.
- Extract reference: debt interest/tax revenue at highest level since 1980s — crowding out other spending.
- Quantitative tightening crystallises losses on the APF, further pressuring the fiscal position via the Treasury indemnity.
- Risk premium effect — higher yields may reflect higher perceived sovereign risk, compounding costs.
- Automatic stabilisers complicate: slower growth from higher yields reduces tax receipts and widens the deficit.

Diagram

- Bond price / yield diagram with inverse relationship; show original equilibrium (high price, low yield) and new equilibrium (lower price, higher yield).
- Alternative acceptable: loanable funds model — rightward shift in demand for loanable funds from government borrowing, raising the equilibrium real interest rate.

Marking notes

- Cap at 5/8 without a correct diagram or with a seriously flawed one.
- Top-band answers must link yields → debt servicing costs → fiscal position quantitatively, not just describe.

Question 2(c) — Discuss (12 marks)

Microeconomic and macroeconomic effects on UK firms and households of a sustained rise in long-term interest rates.

Marks breakdown

AO1 (K)	AO2 (App)	AO3 (An)	AO4 (Ev)
3	3	3	3

Microeconomic effects

- Firms: higher cost of capital reduces NPV of long-dated investment projects — capital-intensive sectors (utilities, housebuilders, infrastructure) most affected.
- Small firms reliant on bank credit face higher refinancing costs; credit rationing tightens.
- Equity valuations fall as discount rates rise — reduces incentive for equity-financed scale-up investment.
- Households: higher mortgage rates reduce disposable income; long-term savers benefit.
- Pension funds: higher yields actually improve funding ratios of DB schemes — one positive distributional effect.

Macroeconomic effects

- AD: lower I and C → AD shifts left; multiplier effect on employment.
- LRAS: sustained weak investment reduces the growth rate of LRAS, potential output falls.
- Exchange rate: higher UK yields attract capital inflows, appreciating sterling — lower import prices but weaker export competitiveness.
- Fiscal position: higher debt servicing costs crowd out public investment — further LRAS damage.
- Inflation: AD effect is disinflationary, but fiscal-dominance risks (monetary financing pressure) could be inflationary over the long run.

Evaluation points expected

- Heterogeneity: winners (savers, creditors, DB pension holders) and losers (mortgage-holders, debtors, indebted firms).
- Depends on whether the rise is cyclical or structural (e.g. permanent rise in neutral real rate).
- Depends on the cause — higher growth expectations (benign) vs higher risk premium (malign).
- Policy interactions: monetary and fiscal policy must be consistent to avoid further yield pressure.
- Overall: likely contractionary in the short run and a structural headwind in the long run unless productivity responds.

Question 2(d) — Evaluate (25 marks)

Micro and macro effects on the UK economy of a sustained rise in the 10-year gilt yield above 5%.

Marks breakdown

AO1 (K)	AO2 (App)	AO3 (An)	AO4 (Ev)
4	6	7	8

Indicative content — microeconomic effects

- Firm-level cost of capital rises sharply — projects with IRR between 4% and 5%+risk premium are no longer viable.
- Sectoral impact: housebuilders, commercial real estate, infrastructure, utilities, early-stage tech disproportionately affected.
- Household balance sheets: mortgage rates track gilt yields; refinancing shock for the ~30% of owner-occupiers with fixed-rate deals maturing.
- Commercial banks: net interest margins rise but loan-loss provisions also rise; smaller banks may face deposit outflows.
- Insurers and pension funds: improved funding ratios but mark-to-market losses on existing gilt holdings (LDI-style stress risks).
- Start-ups and scale-ups: higher discount rates collapse valuations — funding winter.

Indicative content — macroeconomic effects

- AD: lower consumption and investment — AD shifts left; growth slows, output gap widens.
- Unemployment: rises, especially in interest-sensitive sectors.
- Inflation: short-run disinflation through AD channel; possible stagflation risk if supply side also weakens.
- Fiscal position: sharp rise in debt interest costs; tight fiscal space forces austerity or higher taxes.
- Crowding out: higher gilt yields raise the hurdle rate for private investment — reduces LRAS growth.
- Exchange rate: sterling likely appreciates initially on rate differential, then depreciates if sovereign risk dominates.
- Debt dynamics: if $r > g$ persistently, debt ratio rises even with a primary balance — snowball effect.
- Financial stability: mortgage defaults, commercial real estate stress, non-bank financial institution fragility.

Diagram suggestions

- AD/AS diagram — AD shift left and possibly LRAS shift left in the long run.
- Loanable funds market — higher equilibrium rate with lower investment.

Evaluation points expected

- Cause matters: structural rise in the global neutral rate (external, cannot fight) vs UK-specific risk premium (policy can address).
- Duration: transient spikes differ from sustained 5%+ — the question specifies “sustained”, so longer-run structural effects dominate.
- Heterogeneity of impact across sectors, generations and regions — distributional politics becomes acute.
- Policy mix: fiscal consolidation, supply-side reform and macroprudential tools can partially offset the macro drag.
- Feedback: monetary policy may cut Bank Rate in response — but long-end yields are driven by term premium and expected future rates, not just the policy rate.

- International comparison — many advanced economies face similar yields, so relative position matters more than absolute level.
- Judgement: likely net negative, with the severity depending on the cause and on fiscal-monetary coordination.

Levels apply (see front of mark scheme)

Question 2(e) — Evaluate (25 marks)

Case for the UK adopting binding fiscal rules to reduce public sector net debt below 80% of GDP by 2035.

Marks breakdown

AO1 (K)	AO2 (App)	AO3 (An)	AO4 (Ev)
4	6	7	8

Indicative content — case for binding rules

- Lower debt builds insurance against future shocks — fiscal space for counter-cyclical policy.
- Lower sovereign risk premium reduces gilt yields, lowering borrowing costs for households and firms (crowding in).
- Anchoring expectations of fiscal sustainability helps the BoE — monetary policy more effective when fiscal dominance risk is low.
- Reduces the risk of debt spirals if r exceeds g .
- Intergenerational fairness — avoids loading costs on future taxpayers.
- International evidence: credible fiscal frameworks (e.g. Swiss debt brake, post-2012 eurozone rules) are associated with lower yields and fewer crises.
- Discipline on time-inconsistent political behaviour — binds government to its long-run interest.

Indicative content — case against binding rules

- Pro-cyclical risk: rules can force tightening during downturns, deepening recessions.
- Public investment is squeezed first (it's the most politically easy to cut), damaging LRAS and long-run growth.
- Ricardian equivalence is only partial; demand-side effects of consolidation are real and negative in the short run.
- When $g > r$ (recent UK history pre-2022), debt-to-GDP can fall even with a primary deficit — rules may be unnecessarily tight.
- Rules are often circumvented (off-balance-sheet liabilities, creative accounting) — credibility may be illusory.
- UK has monetary sovereignty, deep capital markets and reserve-asset status (sterling) — higher debt is more sustainable than for small open economies.
- Hysteresis: consolidation that damages long-run growth can worsen the debt ratio by shrinking the denominator.
- Political economy: binding rules can undermine democratic fiscal choice over a 10-year horizon.

Application expected

- Use of Figure 2 data: debt is near 99%, well above the 80% target — implies a large adjustment.
- Use of extract: “ $g > r$ ” argument, QT, risk premium, debt-interest/tax ratio at highest since 1980s.
- Reference to UK institutional context: OBR, fiscal rules history (Golden Rule, Sustainable Investment Rule, Charter for Budget Responsibility).

Evaluation points expected

- Design matters: flexible rules with escape clauses for shocks outperform rigid rules.
- Composition matters: rules that protect public investment (e.g. current-balance rather than overall-balance targets) avoid damaging LRAS.

- Time horizon: a 10-year target requires consistent policy across several parliaments — political feasibility is a real constraint.
- Alternative approach: independent fiscal institution oversight (OBR) with non-binding targets may deliver similar credibility benefits at lower cost.
- Depends on r and g path: if $r < g$ persists, the case for aggressive consolidation is weaker.
- Interaction with monetary policy — consolidation paired with looser monetary policy minimises the output cost (policy mix).
- Supported judgement: a flexible rules-based framework that protects public investment and uses escape clauses is likely more effective than rigid debt targets; 80% by 2035 may be too ambitious if achieved through consolidation alone, but achievable if growth-enhancing policies succeed.

Levels apply (see front of mark scheme)

TOTAL MARKS AVAILABLE: 100

END OF MARK SCHEME